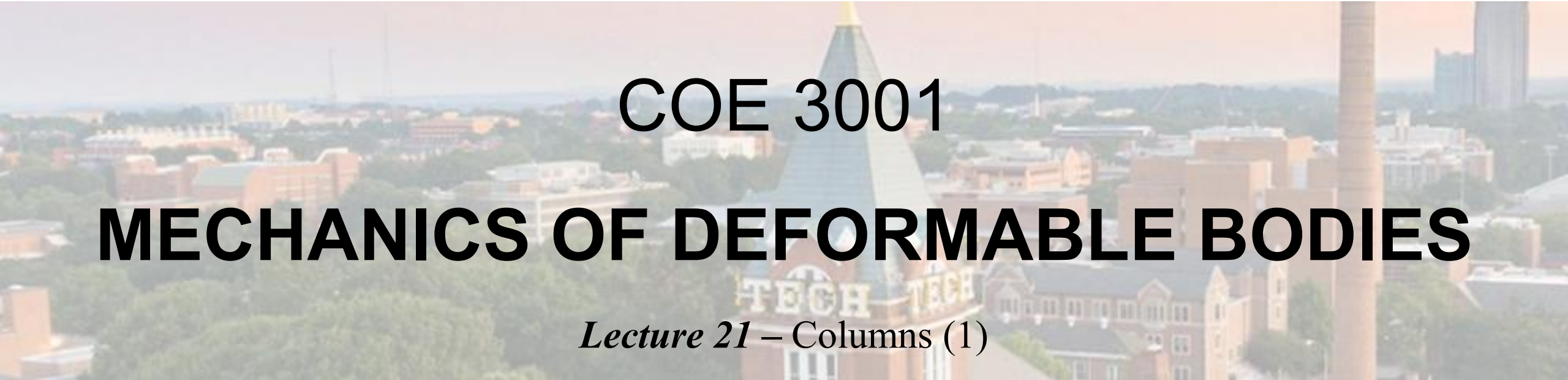




Georgia Tech
College of
Engineering

A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus, showing various buildings and greenery under a soft sky.

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 21 – Columns (1)

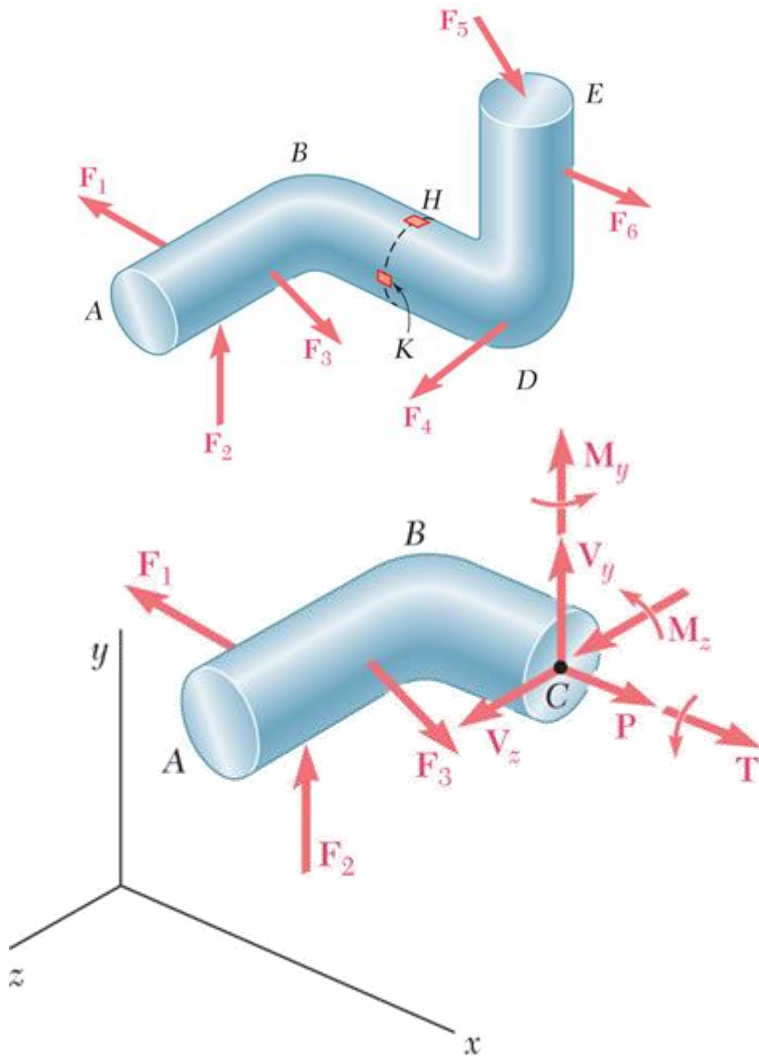
Yazhuo Liu

Georgia Institute of Technology

Apr. 15, 2026

Review of combined load

in general : 3 forces P, V_y, V_z
 3 moments T, M_y, M_z



$$\vec{F}_R = -\sum \vec{F}_i, \quad \vec{M}_R = -\sum \vec{r}_i \times \vec{F}_i$$

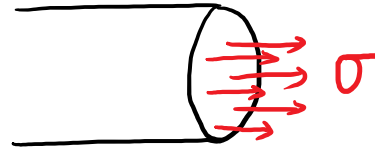
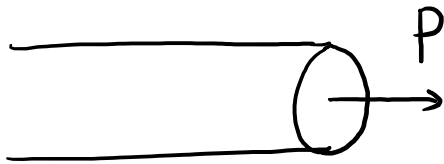
$$\vec{F}_R = P\vec{i} + V_y\vec{j} + V_z\vec{k}$$

$$\vec{F}_i = F_{ix}\vec{i} + F_{iy}\vec{j} + F_{iz}\vec{k}$$

$$\vec{r}_i = a\vec{i} + b\vec{j} + c\vec{k}$$

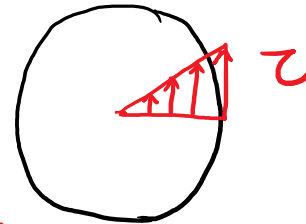
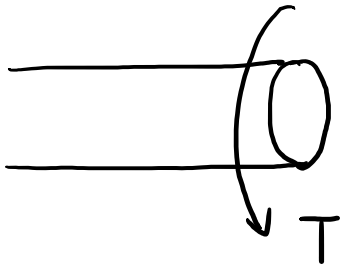
Stress analysis

- Normal force P .



$$\sigma_x = \frac{P}{A}$$

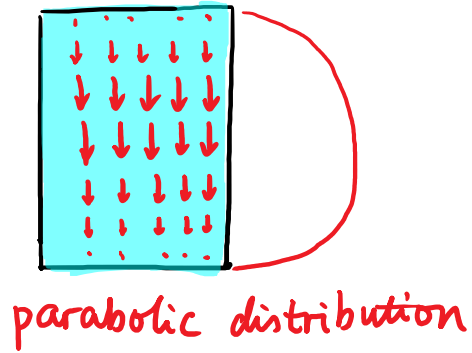
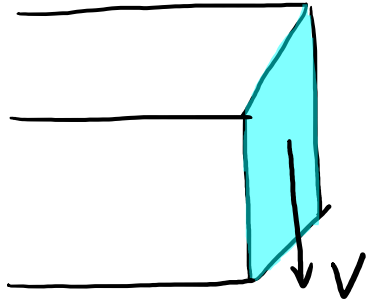
- Torsional moment T . (circular shaft)



linear distribution

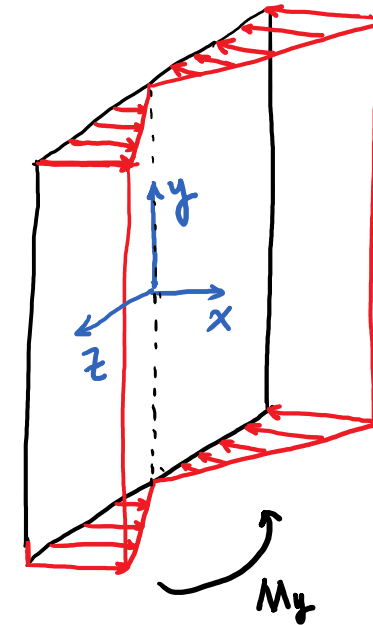
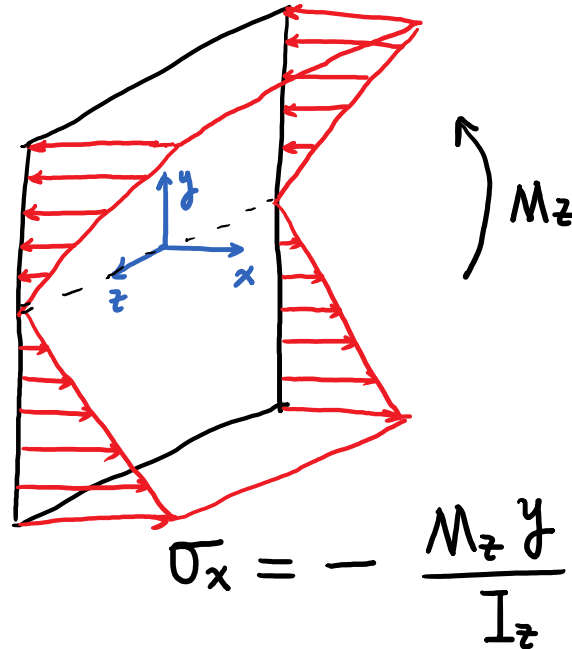
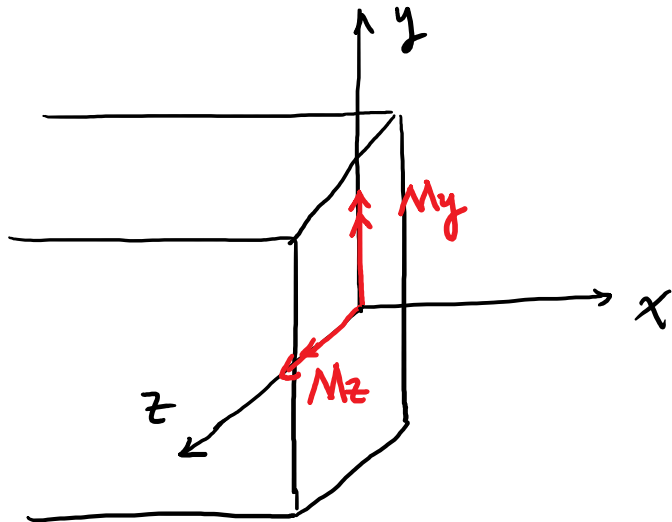
$$\tau = \frac{T\rho}{J}$$

◦ Shear force V



$$\tau = \frac{VQ}{Ib}$$

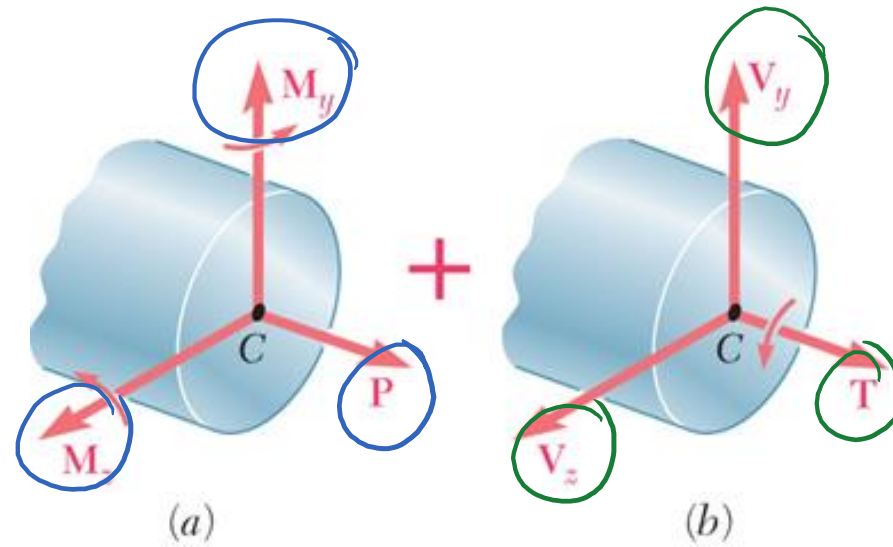
◦ Bending moment M



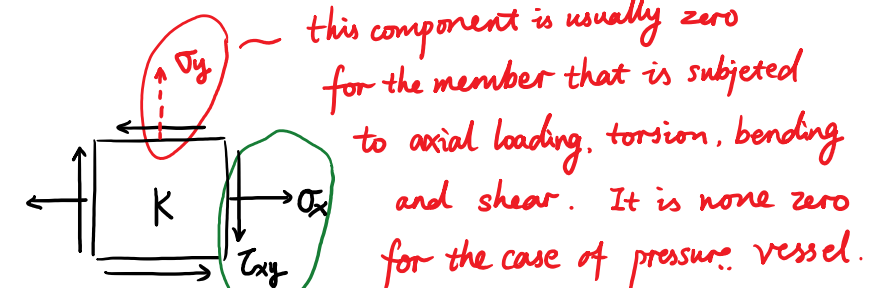
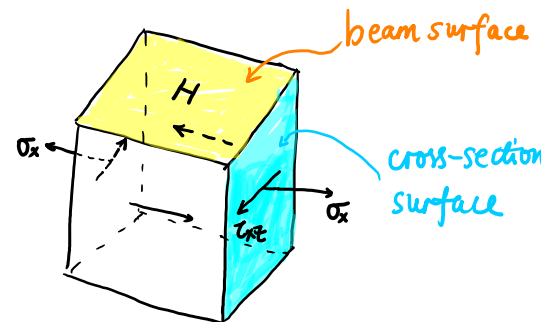
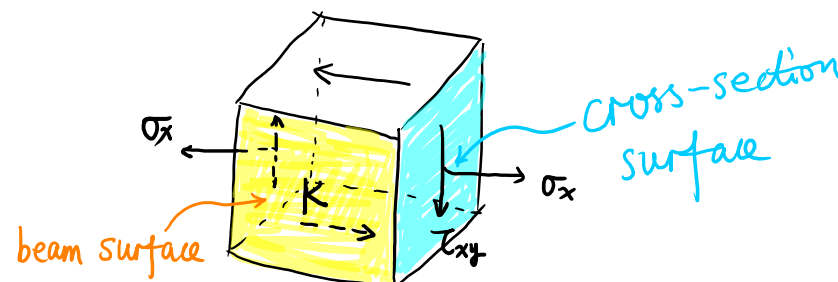
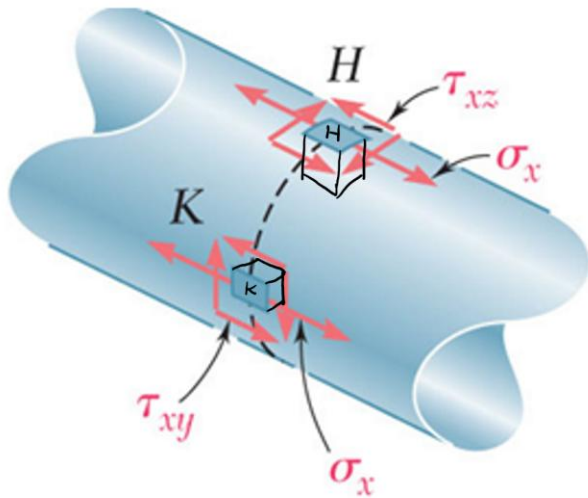
at positive z location
you get tension.

$$\sigma_x = (+) \frac{M_y z}{I_y}$$

axial force and
in-plane moments
 $\Rightarrow$ normal stress



shear force and
axial twisting moment
 $\Rightarrow$ shear stress



These two components are calculated on the cross section surface, and the stress on the other three surface are subsequently drawn.

Strength theories

- Maximum tensile stress theory

$$\sigma_{r1} = \sigma_1 \leq [\sigma] = \frac{\sigma_y}{[n]}$$

- Maximum tensile strain theory

$$\sigma_{r2} = \sigma_1 - \nu(\sigma_2 + \sigma_3) \leq [\sigma] = \frac{\sigma_y}{[n]}$$

- Maximum shear stress theory

$$\sigma_{r3} = \sigma_1 - \sigma_3 \leq [\sigma] = \frac{\sigma_y}{[n]}$$

- Maximum distortional strain energy density theory

$$\sigma_{r4} = \sqrt{\frac{1}{2} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \leq [\sigma] = \frac{\sigma_y}{[n]}$$

Three issues in mechanics of materials

- Strength

- Yielding, fracture

Too much stress

- Stiffness

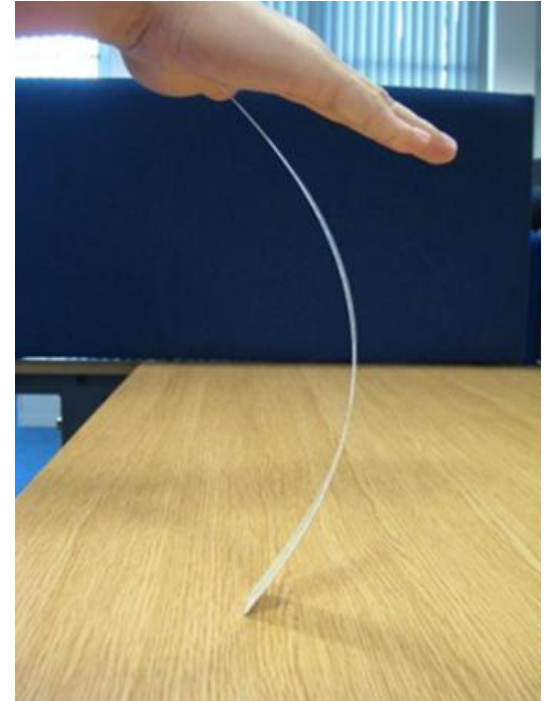
- Too large deformations

Too much strain

- Stability

- Loss of stable configuration

Buckling



Buckling

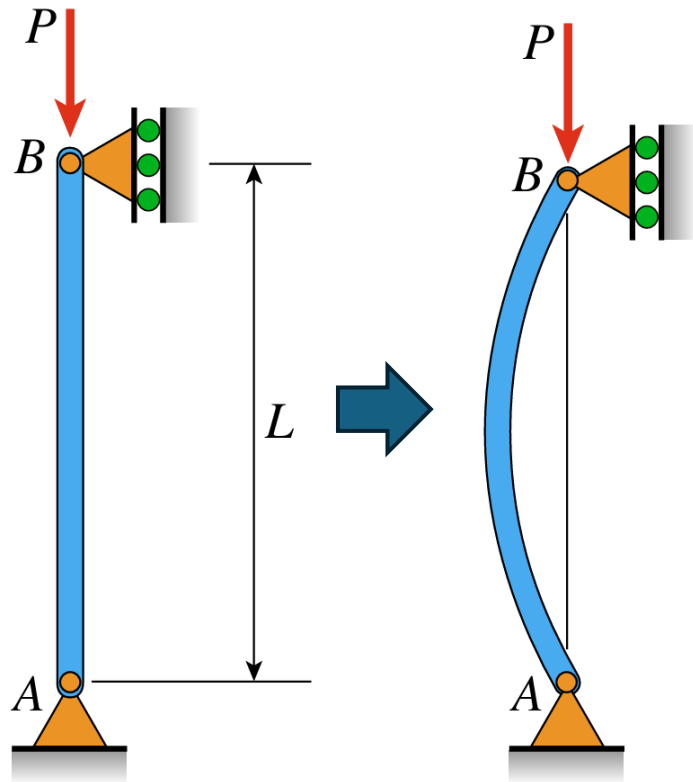
In structural engineering, buckling is the sudden change in shape (deformation) of a structural component under load, such as the bowing of a column under compression.



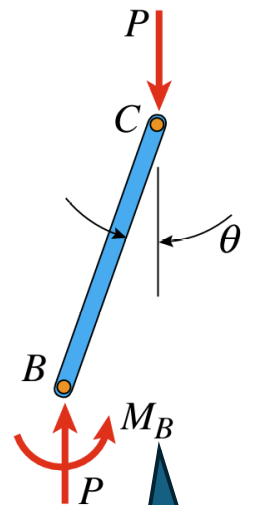
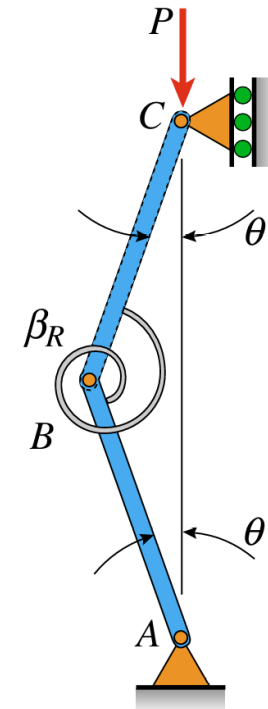
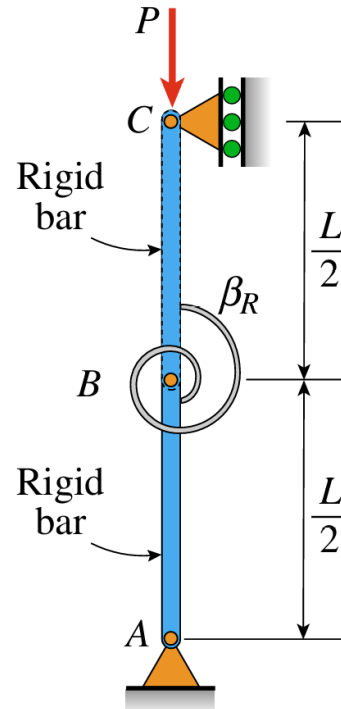
Buckled skin panels on a B-52 aircraft. Thin skin panels buckle at very low loads. In the case shown here, the weight of the forward fuselage structure ahead of the nose undercarriage is sufficient to cause the panels to buckle.

Stability of columns

To illustrate the fundamental concepts of buckling, consider the **idealized structure**, or **buckling model**



Lost stability



restoring
moment

Critical load

The moment M_B is equal to the rotational stiffness β_R times the angle of rotation 2θ of the spring

$$M_B = 2\beta_R\theta$$

Moment balance

$$M_B - P\left(\theta \times \frac{L}{2}\right) = 0$$

Therefore

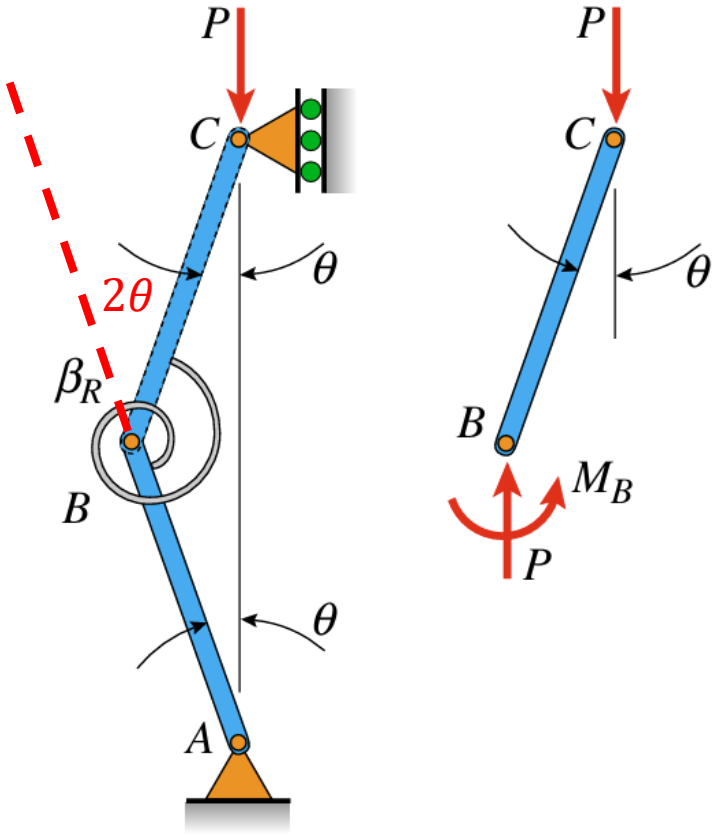
$$\left(2\beta_R - \frac{PL}{2}\right)\theta = 0$$

Solution 1:

$$\theta = 0$$

Solution 2:

$$P_{cr} = \frac{4\beta_R}{L}$$



Critical load

Restoring moment

$$M_B = 2\beta_R \theta$$

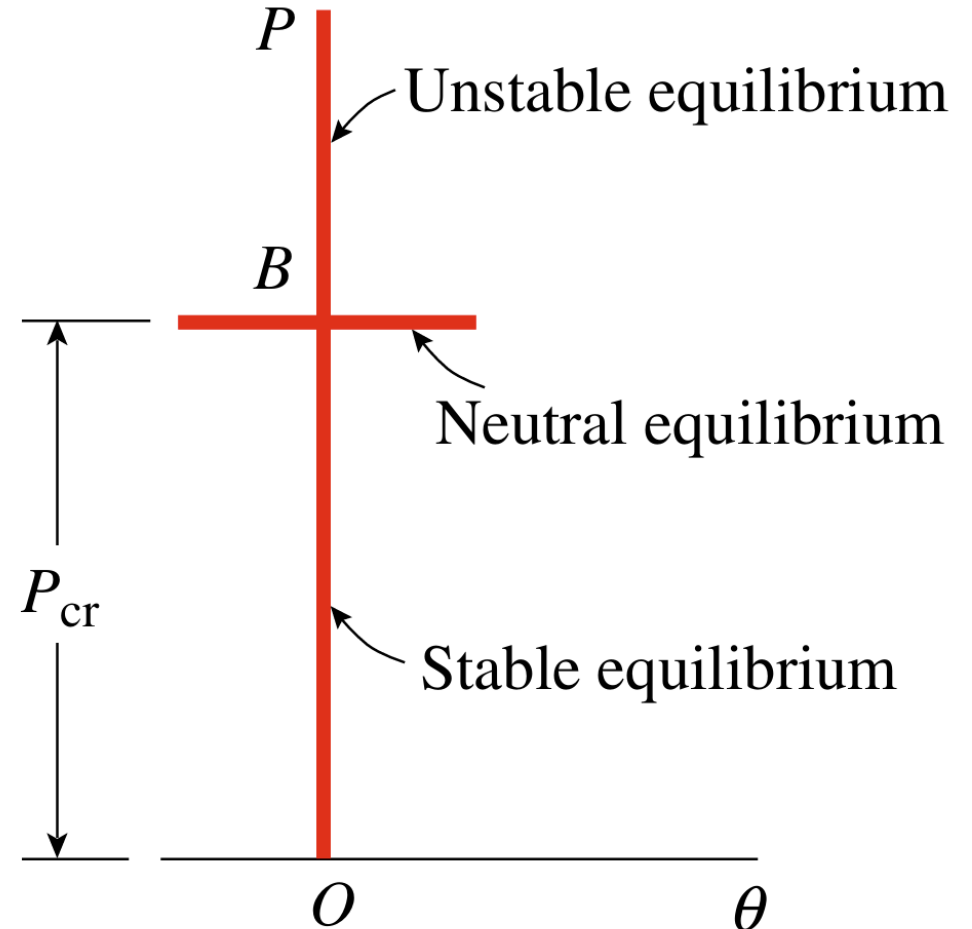
Destabilizing moment

$$P \frac{L}{2} \theta$$

Column is stable (tends to return to aligned orientation) if

$$P \frac{L}{2} \theta < 2\beta_R \theta$$

$$\Rightarrow P < P_{\text{cr}} = \frac{4\beta_R}{L}$$



Buckling behavior

Buckling is a mode of failure under compression of a structural component that is **long and slender** or **thin** (e.g. shell structure).

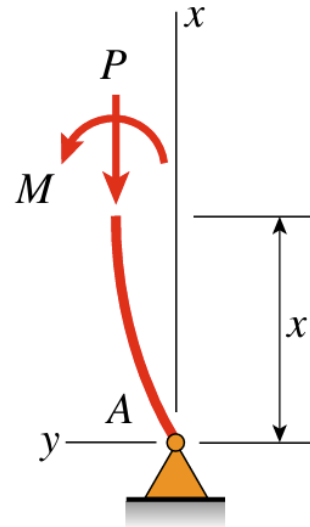
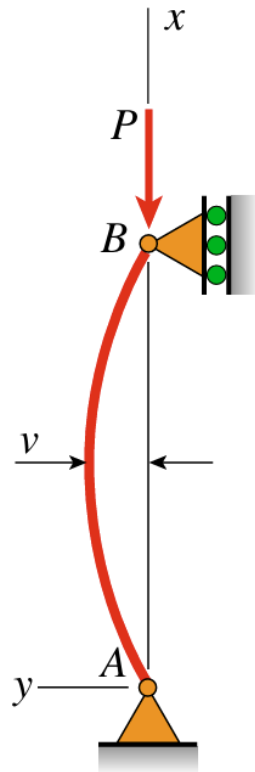
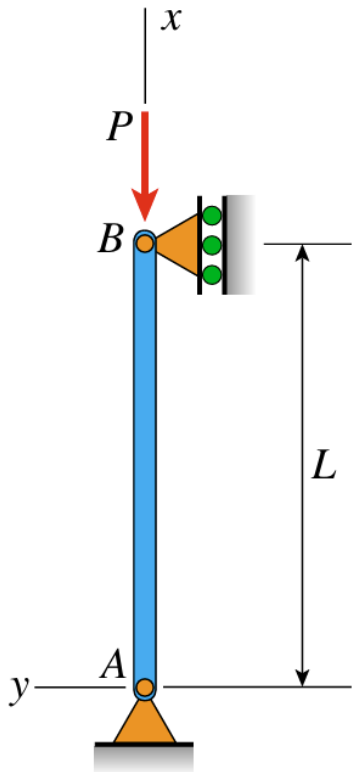
e.g.:

- 1) long, slender members subjected to axial compressive force
- 2) wrinkling in thin-walled structures

Features of buckling:

- 1) occurring only in compression and above a critical level.
- 2) failure by large lateral deflection
- 3) materials are still safe on strength

Differential equation for column buckling



For an ideal pin-ended column

$$EIv'' = M$$

Moment equilibrium

$$M = -Pv$$

Gives

$$EIv'' + Pv = 0$$

Solution of the differential equation

$$EIv'' + Pv = 0$$

Let $k^2 = P/EI$, then

$$v'' + k^2v = 0$$

Seek $v = e^{rx}$ for the solution

$$r^2e^{rx} + k^2e^{rx} = 0 \Rightarrow r = \pm ik$$

The general solution

$$v(x) = C_1e^{ikx} + C_2e^{-ikx}$$

In real case

$$v(x) = A \sin kx + B \cos kx$$

Boundary conditions

$$v(0) = 0 \Rightarrow B = 0$$

$$v(L) = 0$$

$$\Rightarrow A = 0$$

$$\Rightarrow \sin kL = 0$$

$$\Rightarrow kL = 0, \pi, 2\pi, \dots$$

Not
curved

Euler's formula for pin-ended beams

$$\sqrt{\frac{P}{EI}} L = 0, \pi, 2\pi, \dots$$

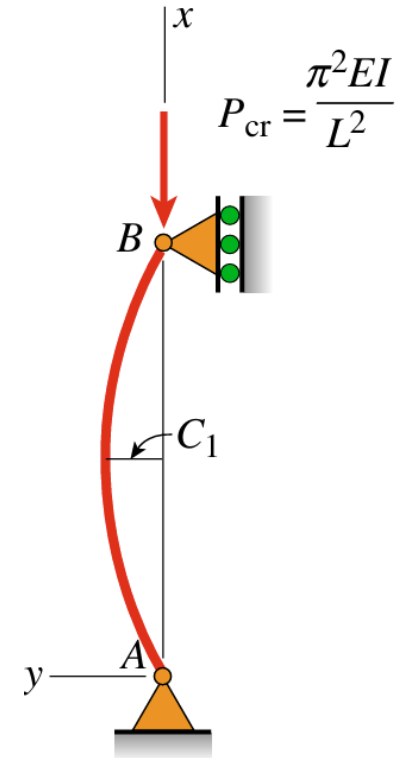
When $p \neq 0$, the minimal solution $kL = \pi$ corresponds to the minimum critical force, which is the engineering critical force

$$\sqrt{\frac{P_{\text{cr}}}{EI}} L = \pi \Rightarrow P_{\text{cr}} = \frac{\pi^2 EI}{L^2}$$

$$v(x) = A \sin \frac{\pi x}{L}$$

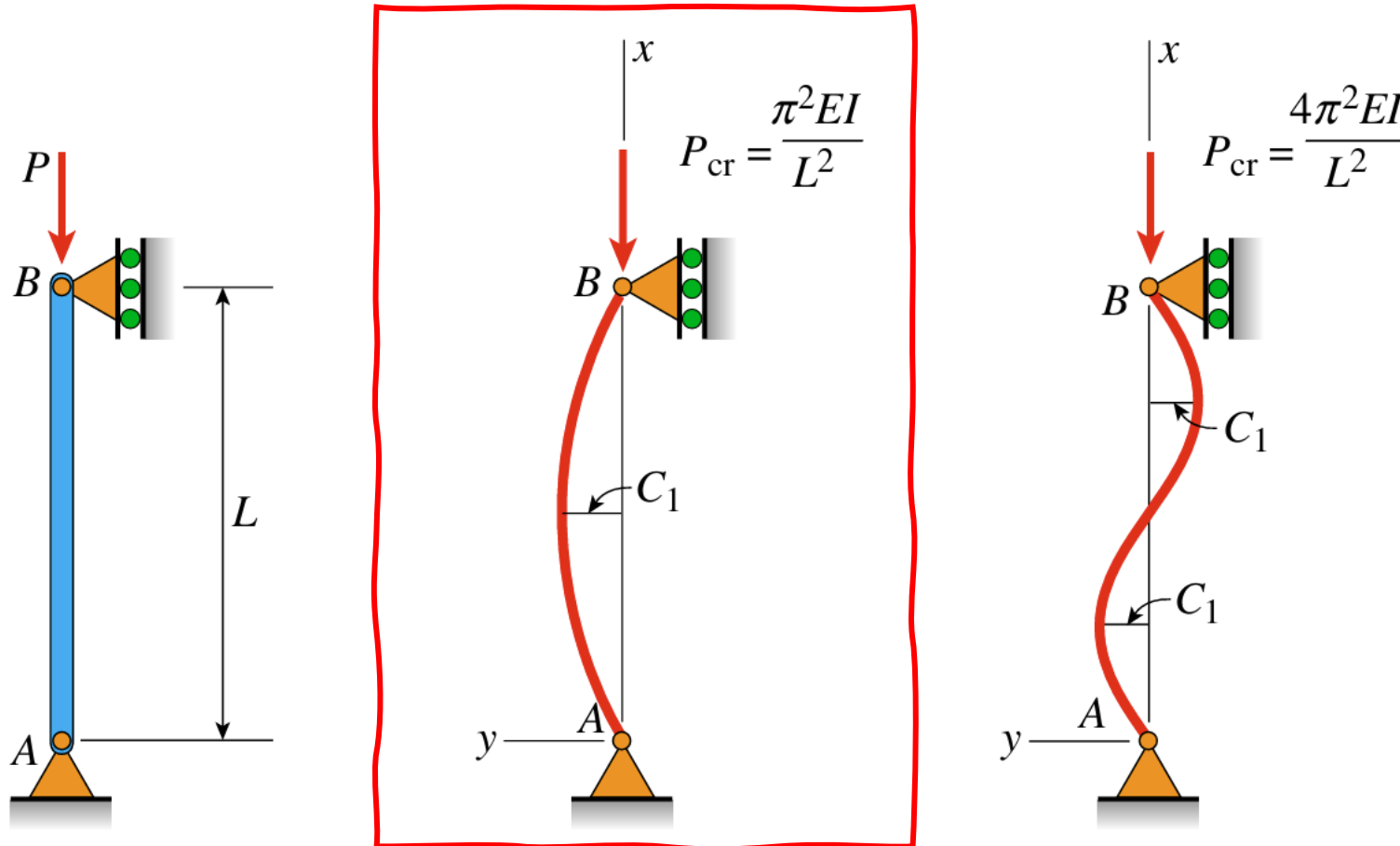
Euler's formula for
pin-ended beams

Critical deflection
curve



Euler buckling

Buckling of a pinned-end column in the first mode is called **the fundamental case of column buckling**.



Critical stress

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 EI}{AL^2}$$

This equation can be written in a more useful form by introducing the notation

$$r = \sqrt{\frac{I}{A}} \Rightarrow \sigma_{cr} = \frac{\pi^2 E}{(L/r)^2}$$

Slenderness ratio

$$\lambda = L/r$$

